

The Value Premium

Lakonishok, Shleifer & Vishny, Fama–French, and Daniel–Titman —
is the premium risk, or is it mispricing?

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MGT 595 · Quantitative Investing

Lecture 4

What this deck covers

One effect, three landmark papers, and a debate.

- **Lakonishok, Shleifer & Vishny (1994)** — *Contrarian Investment, Extrapolation, and Risk*
The value/growth effect, behavioral lens: extrapolation and overreaction, not risk
- **Fama & French (1996)** — *Multifactor Explanations of Asset Pricing Anomalies*
One three-factor model absorbs (almost) every anomaly; a rational, risk-based explanation for value
- **Daniel & Titman (1997)** — *Characteristics vs. Covariances*
An attempt to break the logjam: do covariances price returns, or do characteristics?
- **The replies that followed**
Davis, Fama & French (2000) and Berk (2001) push back on the Daniel–Titman test

The question running through all of it

Value stocks outperform growth stocks — dramatically and persistently. Is that a *risk premium* you earn for bearing distress, or a *mistake* the market keeps making? Your answer determines whether the premium is risk-based or behavioral.

And, as an investor, do you — or should you — care?

Lakonishok, Shleifer & Vishny (1994)

Value and growth: the basic sort

A value stock is cheap relative to fundamentals; a growth stock is expensive.

Forming the portfolios

A **value** stock has a *high* ratio of fundamentals to price:

Book-to-market BE/ME

Cash flow-to-price C/P

Earnings-to-price E/P

A **growth** (“glamour”) stock has *low* values of these ratios — the market pays up for it.

The stylized fact

Sort on any of these ratios and the cheap portfolio *dramatically* outperforms the expensive one — on the order of 10% per year, before any risk adjustment.

The whole lecture in one question

value beats growth
not in dispute

$\Rightarrow$

why?
the entire debate

LSV (1994): investors *extrapolate* too far; the premium is a behavioral error, not compensation for risk

FF (1996): value stocks load on a *distress* factor; the premium is a rational risk price

DT (1997): try to resolve this debate through a clever test

LSV's thesis: extrapolation and overreaction

Contrarian strategies work by trading against naive investor behavior — and they are not riskier.

The behavioral story

Glamour: stocks that have done well become exciting. Investors stampede in, push prices up, and *depress* future returns.

Value: neglected, poorly performing firms get overly punished. Prices fall too far, and expected returns *rise*.

This is an *overreaction* story — people extrapolate recent performance too far into the future.

A contrarian strategy simply takes the other side.

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Where the mistake comes from

Investors extrapolate earnings, overreact to news, assume trends persist, and equate a *good company* with a *good investment*. Psychological evidence says these naive heuristics are exactly what people use.

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And not riskier — LSV's four claims

Value stocks do *not* have:

- higher variance
- higher betas
- more downside risk
- underperformance in bad states

Making the story testable

LSV turn “extrapolation” and “not riskier” into two things you can measure.

1. Past vs. expected future performance

Glamour stocks should have *good* past performance *and* good expected future performance; value stocks, bad and bad. The two are distinct and separately measurable:

Past: realized growth in sales, earnings, cash flow

Expected future: the price multiples E/P , C/P — what the market is paying *for* growth

2. Assessing risk properly

For the premium to be a risk premium, value must underperform glamour:

with some *frequency*, and

in states of the world that *matter* to investors — low-consumption, high marginal-utility states.

A premium earned in good times only is not compensation for risk.

The design

Separate *past* from *expected future* growth, then check whether the return spread lines up with extrapolation errors — or with bad-state risk.

A clean long-horizon design — with the selection-bias traps flagged up front.

The sample

NYSE-AMEX firms, 1963–1990 (CRSP returns + COMPUSTAT accounting)

Strategies start in 1968 — 5 years of past history are required first

Form at t ; hold and compound annually over $t+1$ to $t+5$ (equal-weighted)

Focus on *long-term buy-and-hold* returns

Data and methodology

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Handling disappearing firms

If a firm delists mid-year, replace its return to year-end with a same-size-decile portfolio. *What return should delisted firms get — and does this bite value or growth more?*

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Selection-bias questions to keep asking

COMPUSTAT *back-fills* data — survivors look better than they were. Which way does this push value vs. growth?

Delisting returns are easy to mis-assign, and failures cluster in distressed (value) names.

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Two deliberate choices

Adjust for size with *characteristic-matched* benchmarks, not regressions

Build growth rates and growth *expectations* from accounting ratios

Measuring growth without letting it explode

Firm-level growth rates are wildly volatile and undefined when earnings go negative. The fix: average first, then divide.

The naive portfolio earnings yield

Invest \$1 in each of n firms at $t-4$; aggregate earnings relative to market cap:

$$E_{p,t-4} = \frac{1}{n} \sum_{i=1}^n E_{i,t-4} \frac{\$1}{\text{Mkt.Cap.}_{i,t-4}}$$

A one-year growth rate

$g_{t-4:t-3} = (E_{p,t-3} - E_{p,t-4})/E_{p,t-4}$ is very noisy — and blows up when $E_{p,t-4} < 0$.

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Average over time first

$$AE_{p,t-4} = \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{n} \sum_{i=1}^n E_{i,t-4} \frac{\$1}{\text{Mkt.Cap.}_{i,t-4}} \right)$$

$$g_{t-4:t-3} = \frac{AE_{p,t-3} - AE_{p,t-4}}{AE_{p,t-4}}$$

Smooths the noise and keeps the denominator sensible.

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Smooths the noise and keeps the denominator sensible.

Also use sales growth

Past sales growth is steadier than earnings or cash flow and is not contaminated by price.

Expected future growth, via the Gordon growth model

A low profitability-to-price ratio is the market betting on high future growth.

The intuition

From the Gordon growth model,

$$P = \frac{D}{r - g}.$$

Hold r roughly fixed: a *high* price relative to fundamentals ($\Rightarrow$ low C/P , E/P) means the market expects *high* g . Profitability-to-price ratios proxy for expected growth.

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Why split past from future at all

The breakdown tells us *what a low BE/ME really is*:

a firm with intangible assets,
genuine growth opportunities,
a low-risk, low-discount-rate stock, or
an *overvalued glamour* stock.

LSV want to know which — and the past/future split is how they find out.

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An alternative measure

One could also use analysts' forecasts of future growth (La Porta 1993) instead of multiples.

The refined contrarian strategy

Don't just sort on cheapness — sort on cheapness and on the growth investors are extrapolating.

Double sort

Glamour: high past growth (high *GS*) and high expected growth (low *C/P*)

Value: low past growth and low expected growth (high *C/P*)

Buy low-past, low-future; sell high-past, high-future

	Glamour						Value		
	1	1	①	2	2	2	③	3	3
Future = C/P	1	2	②	1	2	3	①	2	3
Past = GS	1	2	③	1	2	3	②	1	3
$\Rightarrow R_1$	0.157	0.131	0.113	0.181	0.156	0.139	0.215	0.202	0.137
R_2	0.147	0.120	0.100	0.191	0.165	0.167	0.213	0.188	0.165
R_3	0.165	0.140	0.121	0.197	0.190	0.165	0.227	0.195	0.172
R_4	0.164	0.124	0.114	0.198	0.169	0.166	0.231	0.204	0.177
R_5	0.179	0.135	0.121	0.200	0.173	0.151	0.218	0.216	0.184
AR	0.162	0.130	0.114	0.193	0.171	0.157	0.221	0.201	0.167
CR_1	1.122	0.843	0.712	1.419	1.200	1.076	1.711	1.497	1.163
$SAAR$	-0.006	-0.020	-0.033	0.030	0.014	0.003	0.054	0.036	0.008

Table II, Panel A: $C/P \times GS$ double sort, annual buy-and-hold returns.

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Panel A: C/P and GS									
Future = C/P Past = GS	Glamour			Value					
	1	1	①	2	2	2	③	3	3
	1	2	②	1	2	3	①	2	3
⇒ R_1	0.157	0.131	0.113	0.181	0.156	0.139	0.215	0.202	0.137
R_2	0.147	0.120	0.100	0.191	0.165	0.167	0.213	0.188	0.165
R_3	0.165	0.140	0.121	0.197	0.190	0.165	0.227	0.195	0.172
R_4	0.164	0.124	0.114	0.198	0.169	0.166	0.231	0.204	0.177
R_5	0.179	0.135	0.121	0.200	0.173	0.151	0.218	0.216	0.184
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Table II, Panel A: $C/P \times GS$ double sort, annual buy-and-hold returns.

Read the two circled cells

The circled glamour and value cells (R_1) are 11.3% and 21.5% — a value premium of 10.2 points, far wider than a one-way *C/P* sort. Sorting on *both* past and expected-future growth maximizes the gap.

The spread is everywhere: every value measure

Table II (cont.): the gap survives when “cheap” is E/P or BE/ME instead of C/P .

Future Past	Glamour						Value		
	1	1	1	2	2	2	3	3	
	1	2	3	1	2	3	1	2	3
R_1	0.184	0.148	0.118	0.188	0.153	0.139	0.224	0.205	0.174
R_2	0.167	0.134	0.100	0.204	0.174	0.154	0.214	0.187	0.190
R_3	0.185	0.153	0.119	0.222	0.189	0.169	0.221	0.198	0.189
R_4	0.190	0.138	0.103	0.205	0.175	0.160	0.232	0.217	0.188
R_5	0.189	0.163	0.104	0.201	0.180	0.157	0.215	0.210	0.199
AR	0.183	0.147	0.109	0.204	0.174	0.156	0.221	0.203	0.188
CR_5	1.315	0.986	0.674	1.533	1.230	1.063	1.716	1.523	1.365
SAAR	0.005	-0.011	-0.037	0.033	0.013	0.002	0.040	0.034	0.017

Panel C: B/M and GS									
Future Past	Glamour						Value		
	1	1	1	2	2	2	3	3	3
	1	2	3	1	2	3	1	2	3
R_1	0.147	0.141	0.132	0.160	0.159	0.121	0.204	0.185	0.135
R_2	0.127	0.138	0.127	0.175	0.166	0.150	0.200	0.172	0.163
R_3	0.149	0.149	0.137	0.190	0.186	0.152	0.221	0.192	0.182
R_4	0.147	0.130	0.130	0.191	0.176	0.154	0.222	0.190	0.195
R_5	0.158	0.140	0.124	0.203	0.180	0.165	0.216	0.211	0.164
AR	0.146	0.140	0.130	0.184	0.173	0.148	0.212	0.190	0.168
CR_5	0.974	0.925	0.842	1.325	1.224	0.996	1.618	1.387	1.171
SAAR	-0.009	-0.012	-0.021	0.022	0.015	-0.009	0.039	0.030	0.017

Circled cells, value – glamour (R_1)

Panel B, $E/P \times GS$: $22.4\% - 11.8\% = 10.6$ pts. Panel C, $BE/ME \times GS$: $20.4\% - 13.2\% = 7.2$ pts.

And it holds without a past-growth axis at all

Table II (cont.): two value ratios sorted against each other — no GS — still separate glamour from value.

	Glamour								
	1	1	1	2	2	2	3	3	3
Future = E/P	1	2	3	1	2	3	1	2	3
Past = B/M	1	2	3	1	2	3	1	2	3
R_1	0.116	0.118	0.186	0.142	0.143	0.174	0.135	0.174	0.189
R_2	0.086	0.120	0.194	0.146	0.163	0.192	0.173	0.178	0.185
R_3	0.114	0.154	0.201	0.157	0.184	0.220	0.177	0.178	0.204
R_4	0.093	0.151	0.218	0.150	0.166	0.193	0.188	0.200	0.214
R_5	0.093	0.188	0.218	0.168	0.169	0.209	0.241	0.205	0.204
AR	0.100	0.146	0.203	0.152	0.165	0.198	0.153	0.187	0.199
CR_5	0.613	0.976	1.521	1.032	1.146	1.464	1.311	1.354	1.479
SAAR	-0.039	-0.009	0.022	0.002	0.009	0.033	0.003	0.023	0.030

Panel E: B/M and C/P									
	Glamour								Value
B/M	1	1	1	2	2	2	3	3	3
C/P	1	2	3	1	2	3	1	2	3
R_1	0.111	0.153	0.141	0.101	0.144	0.171	0.170	0.161	0.194
R_2	0.085	0.164	0.172	0.111	0.160	0.181	0.174	0.173	0.189
R_3	0.111	0.172	0.179	0.147	0.177	0.191	0.192	0.206	0.207
R_4	0.101	0.153	0.187	0.155	0.168	0.200	0.177	0.195	0.219
R_5	0.108	0.162	0.250	0.184	0.178	0.208	0.233	0.201	0.209
AR	0.103	0.161	0.186	0.139	0.165	0.190	0.189	0.187	0.203
CR_5	0.633	1.108	1.339	0.917	1.148	1.387	1.378	1.355	1.524

Circled cells, value – glamour (R_1)

Panel D, $E/P \times BE/ME$: 18.9% – 11.6% = 7.3 pts. Panel E, $BE/ME \times C/P$: 19.4% – 11.1% = 8.3 pts.

Robustness check

Before believing it, rule out the usual suspects.

The worries

Just small stocks? The effect survives within size groups.

Selection bias? The back-fill and delisting concerns push, if anything, *against* finding a value premium.

Sort-specific? They run Fama–MacBeth cross-sectional regressions alongside the portfolio sorts.

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Regressions vs. sorts

Fama–MacBeth (Table IV) lets several characteristics compete at once and gives standard errors — but imposes linearity and is sensitive to outliers.

Portfolio sorts are non-parametric and transparent — but coarse and hard to compare across multiple dimensions.

Both point the same way: C/P and past growth (GS) carry independent, significant power.

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Bottom line

The premium is not an artifact of size, survivorship, or one particular methodology.

Robustness check: the Fama–MacBeth regressions

Let every characteristic compete month by month — *GS* and *C/P* keep their power.

What survives

C/P: $t \approx 3.7$ – 4.6 , strongly positive.

GS: $t \approx -2.1$ to -2.8 , high past growth
 $\Rightarrow$ low future return.

BE/ME, *SIZE*: weak or insignificant once
C/P and *GS* are in.

Bottom line

Cash-flow yield and past growth carry
independent information — the same
message as the double sorts.

	Int.	(GS)	B/M	SIZE	E/P+	DE/P	(C/P+)	DC/P
Mean	0.180	-0.061						
t-statistic	3.251	-2.200						
Mean	0.108		0.039					
t-statistic	2.167		2.132					
Mean	0.185			-0.009				
t-statistic	2.140			-1.095				
Mean	0.110				0.526			
t-statistic	2.029				2.541			
Mean	0.099						0.356	
t-statistic	1.873						4.240	
Mean	0.129	-0.058	0.006				0.301	-0.029
t-statistic	2.584	-2.832	<u>0.330</u>				3.697	-1.222
Mean	0.143		0.009	-0.009			0.280	-0.032
t-statistic	1.562		0.565	-1.148			4.223	-1.625
Mean	0.169	-0.044	0.000	-0.009			0.296	-0.036
t-statistic	1.947	-2.125	0.005	-1.062			4.553	-1.625
Mean	0.172	-0.051	0.016	-0.009	0.394	-0.032		
t-statistic	1.961	-2.527	1.036	-1.065	2.008	-1.940		

Table IV: FM cross-sectional regressions. Circles (*GS*, *C/P*) are

Toby's notes.

More evidence: expectations are wrong

If it's extrapolation, then the growth the market priced in should fail to materialize.

Actual vs. expected growth (Table V)

LSV compare *realized* growth after formation to the growth implied by price ratios at formation:

Glamour firms' high expected growth *does not persist* — their advantage fades over the next 5 years.

In years 1–2 expectations look roughly right; by years 3–5 the persistence the market priced in has evaporated.

Exactly the pattern overreaction predicts.

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A slow-motion correction

These are *long-horizon* mistakes that unwind gradually as investors abandon their beliefs — the mirror image of momentum, which is short-horizon *underreaction*.

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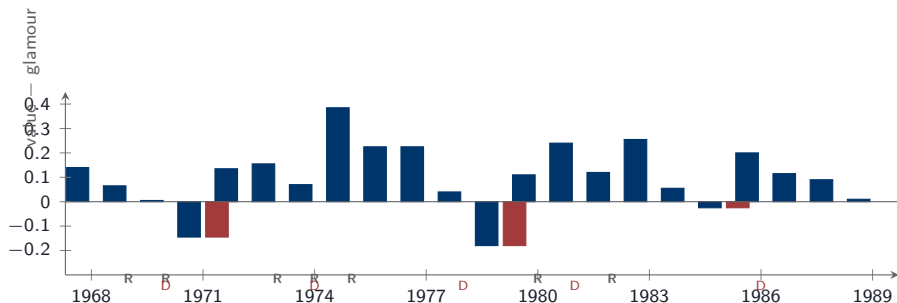
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Why it survives

The correction takes years. Most investors — and most institutions — don't have the horizon to wait it out.

And value wins when it counts

Value minus glamour, year by year, 1968–1989 — positive almost everywhere, including the bad years.



After LSV (1994), Figure 2. **R** = NBER recession; **D** = year the CRSP equal-weighted index fell.

Punchline

Value beats glamour in ≥ 17 of 22 years — and in every 5-year window.

So is it risk? The evidence says no

For the premium to be risk, value has to hurt in bad states. It doesn't.

What LSV find

Value beats glamour in 17/22 years and every 5-year period — including recessions and down markets.

Betas and standard deviations are *not* higher for value.

No tendency to underperform precisely when marginal utility is high.

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The test is motivated by the **ICAPM**: does value covary with bad states (low consumption / wealth)?

But the **APT** is also fair game — maybe value loads on some priced common factor LSV haven't named. That door is left open (and FF96 walk through it).

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So why does it persist?

If it isn't risk, an arbitrageur should compete it away. Perhaps it will be — but until now, no one knew how.

Why a “free” premium can survive

Limits to arbitrage, not the absence of a mistake.

Three frictions

Individuals can't easily overcome the bias — a good company keeps getting confused with a good investment.

Institutions prefer glamour: it is easy to justify to clients.

Horizons are short on both sides; contrarian trades take years to converge.

The honest caveat: data mining

Find a pattern in one 22-year US sample and you should worry you fit noise. The defenses are *out-of-sample*:

international evidence

pre-sample (earlier US) evidence

If the effect shows up where you didn't look for it, it's more likely real.

LSV's verdict

The value premium is a behavioral mistake that survives because it is hard and slow to arbitrage — *not* a reward for risk.

**Fama & French
(1996): one model
for (almost)
everything**

The three-factor model

Add two factors to the market, and a host of CAPM anomalies line up — all but one.

$$\mathbb{E}[\tilde{r}_i] = r_f + b_i(\mathbb{E}[\tilde{R}_M] - r_f) + s_i \mathbb{E}[\text{SMB}] + h_i \mathbb{E}[\text{HML}]$$

estimated as $\tilde{r}_i - r_f = \alpha_i + b_i(\tilde{R}_M - r_f) + s_i \text{SMB} + h_i \text{HML} + \varepsilon_i$

What the factors are

SMB — small minus big (a size factor)

HML — high minus low BE/ME (a value factor)

Loadings s_i , h_i capture common variation the market factor misses

The claim

Many CAPM “anomalies” are *related* and share common variation. One model with these three factors absorbs them — consistent with the **ICAPM** or **APT**. The lone holdout is short-term *momentum*.

HML as a distress factor

Fama and French read the value premium as a rational price for distress risk.

The rational interpretation

Weak, low-earnings firms have *high* h_i . HML captures the common variation running through these distressed names.

Value stocks are typically *fallen angels* — more exposed to this distress source than growth stocks.

More exposure to a priced risk $\Rightarrow$ higher average returns.
The premium is a *wage*, not a free lunch.

SMB, in parallel

SMB captures common variation among small stocks that the market factor leaves on the table.

What FF set out to show

FF (1993) established that this model prices the size and BE/ME portfolios. FF (1996) goes further: it also captures the E/P , C/P , and sales-growth sorts (the LSV portfolios) *and* the DeBondt–Thaler long-term reversal.

HML does most of the work — so the whole interpretation rests on HML.

It explains the value sorts

Three-factor alphas across BE/ME, E/P, C/P, and past-return deciles are economically tiny.

		Deciles										GRS	p(GRS)
		1	2	3	4	5	6	7	8	9	10		
→ BE/ME	Low										High		
	α	0.08	-0.02	-0.09	-0.11	-0.08	-0.03	0.01	-0.04	0.03	-0.00		
	$t(a)$	1.19	-0.26	-1.25	-1.39	-1.16	-0.40	0.15	-0.61	0.43	-0.02	0.57	0.841
	R^2	0.95	0.95	0.94	0.93	0.94	0.94	0.94	0.94	0.95	0.89		
	E/P												
E/P	Low										High		
	α	-0.00	-0.07	-0.07	-0.04	-0.03	0.02	0.06	0.09	0.12	0.00		
	$t(a)$	-0.07	-1.07	-0.94	-0.52	-0.43	0.24	1.01	1.46	1.49	0.05	0.84	0.592
	R^2	0.91	0.95	0.94	0.94	0.94	0.94	0.94	0.94	0.92	0.92		
	C/P												
C/P	Low										High		
	α	0.02	-0.08	-0.07	-0.00	-0.04	0.00	0.00	0.05	0.06	0.01		
	b	1.04	1.06	1.08	1.06	1.05	1.04	0.99	1.00	0.98	1.14		
	s	0.45	0.50	0.54	0.51	0.55	0.50	0.53	0.48	0.57	0.92		
	h	-0.39	-0.18	0.07	0.11	0.23	0.31	0.36	0.50	0.67	0.79		
5-Yr SR	$t(a)$	0.22	-1.14	-1.00	-0.04	-0.51	0.00	0.06	0.72	0.92	0.14	0.49	0.898
	$t(b)$	51.45	61.16	62.49	64.15	59.04	61.28	60.02	63.36	58.92	46.49		
	$t(s)$	15.56	20.32	22.11	21.57	21.49	20.72	22.19	21.17	24.13	26.18		
	$t(h)$	-12.03	-6.52	2.56	4.28	7.85	11.40	13.62	19.46	24.88	19.74		
	R^2	0.93	0.95	0.95	0.95	0.94	0.94	0.94	0.94	0.94	0.92		
5-Yr SR	High										Low		
	α	-0.21	-0.06	-0.03	-0.01	-0.04	-0.02	-0.04	0.00	0.04	0.07		
	b	1.16	1.10	1.09	1.03	1.03	1.03	1.00	0.99	0.99	1.02		
	s	0.72	0.56	0.52	0.49	0.52	0.51	0.50	0.57	0.67	0.95		
	h	-0.09	0.09	0.21	0.20	0.24	0.33	0.33	0.36	0.47	0.50		
5-Yr SR	$t(a)$	-2.60	-0.97	-0.49	-0.20	-0.61	-0.25	-0.66	0.07	0.47	0.60	0.87	0.563
	$t(b)$	59.01	70.59	67.65	65.34	56.68	68.89	62.49	54.12	50.08	34.54		
	$t(s)$	25.69	25.11	22.59	21.65	20.15	23.64	21.89	21.65	23.65	22.34		
	$t(h)$	-2.88	3.55	8.05	7.98	8.07	13.63	12.80	12.13	14.78	10.32		
	R^2	0.95	0.95	0.95	0.95	0.95	0.95	0.94	0.93	0.92	0.87		

Read the right-hand columns

Alphas sit near zero (≤ 9 bp/month); $t(a)$ rarely clears 2. GRS fails to reject for E/P ($p=0.59$), C/P ($p=0.90$), and 5-year reversal ($p=0.56$).

FF (1996): three-factor intercepts α , t -stats, and GRS across decile sorts.

It even captures the refined LSV double sorts

The same three factors price the BE/ME, E/P , and C/P portfolios double-sorted on sales rank.

LSV Ports →

	(1-1)	1-2	1-3	2-1	2-2	2-3	(3-1)	3-2	(3-3)	GRS	p (GRS)
BE/ME & Sales Rank											
α	-0.00	0.00	-0.06	-0.19	-0.00	0.00	-0.19	-0.07	0.07		
β	1.10	1.03	1.00	1.12	1.00	0.99	1.17	1.06	1.01		
s	0.49	0.31	0.55	0.63	0.48	0.50	0.87	0.74	0.97		
h	-0.33	-0.14	-0.04	0.31	0.25	0.32	0.75	0.70	0.68		
$t(\alpha)$	-0.10	0.12	-0.57	-2.59	-0.07	0.12	-1.64	-0.94	0.69	1.22	0.284
$t(\beta)$	71.67	67.85	35.65	61.81	67.36	51.00	41.29	54.45	38.46		
$t(s)$	22.30	14.32	13.77	24.42	22.44	18.18	21.36	26.62	25.76		
$t(h)$	-13.19	-5.74	-0.94	10.57	10.33	10.17	16.30	22.31	15.91		
R^2	0.96	0.95	0.86	0.94	0.95	0.92	0.89	0.93	0.89		
E/P & Sales Rank											
α	-0.06	-0.06	0.02	-0.09	0.03	0.06	-0.19	-0.06	0.06		
β	1.11	1.04	1.02	1.11	1.01	0.99	1.13	1.04	1.00		
s	0.46	0.45	0.74	0.58	0.43	0.48	0.82	0.65	0.92		
h	-0.34	-0.12	0.18	0.14	0.25	0.39	0.53	0.58	0.61		
$t(\alpha)$	-0.89	-0.87	0.24	-1.23	0.53	0.81	-2.10	-0.82	0.59	1.06	0.394
$t(\beta)$	62.12	56.09	41.52	58.97	67.48	53.80	51.32	59.05	37.61		
$t(s)$	18.61	17.04	21.07	21.30	20.18	18.13	26.08	25.66	23.98		
$t(h)$	-11.56	-3.86	4.41	4.50	10.46	12.88	14.92	20.49	14.19		
R^2	0.95	0.94	0.90	0.94	0.95	0.92	0.93	0.94	0.89		
C/P & Sales Rank											
α	-0.02	-0.06	-0.02	-0.14	0.00	0.07	-0.17	-0.02	0.04		
β	1.11	1.01	1.02	1.12	1.02	1.00	1.13	1.04	1.00		
s	0.46	0.42	0.72	0.63	0.46	0.53	0.80	0.64	0.92		
h	-0.36	-0.12	0.14	0.17	0.26	0.34	0.62	0.62	0.68		
$t(\alpha)$	-0.27	-1.03	-0.24	-1.93	0.08	0.95	-1.73	-0.34	0.34	1.04	0.405
$t(\beta)$	64.04	65.82	40.20	63.31	67.96	52.28	45.55	58.48	36.63		
$t(s)$	18.37	19.12	19.86	24.77	21.34	19.47	22.57	25.32	23.47		
$t(h)$	-12.71	-4.90	3.42	5.82	10.61	10.84	15.21	21.64	15.40		
R^2	0.95	0.95	0.89	0.95	0.95	0.92	0.91	0.94	0.88		

Read the GRS column

Even on LSV's own refined sorts, the intercepts stay near zero: GRS *fails* to reject for BE/ME ($p=0.28$), E/P ($p=0.39$), and C/P ($p=0.41$). The hardest case is the sales-growth corner — and the model still passes.

The one thing it can't explain: momentum

Long-horizon reversal, captured. Short-term momentum, emphatically not.

	1	2	3	4	5	6	7	8	9	10	GRS	p(GRS)
Portfolio formation months are $t-12$ to $t-2$												
a	-1.15	-0.39	-0.21	-0.22	-0.04	-0.05	0.12	0.21	0.33	0.59		
b	1.14	1.06	1.04	1.02	1.02	1.02	1.02	1.04	1.03	1.10		
s	1.35	0.77	0.66	0.59	0.53	0.48	0.47	0.45	0.51	0.68		
h	0.54	0.35	0.35	0.33	0.32	0.30	0.29	0.23	0.23	0.04		
$t(a)$	-5.34	-3.05	-2.05	-2.81	-0.54	-0.93	1.94	3.08	3.88	4.56	4.45	0.000
$t(b)$	21.31	33.36	42.03	51.48	61.03	73.62	68.96	62.67	51.75	35.25		
$t(s)$	17.64	16.96	18.59	20.87	22.06	23.96	21.53	19.03	16.89	14.84		
$t(h)$	6.21	6.72	8.74	10.18	11.86	13.16	11.88	8.50	6.68	0.70		
R^2	0.75	0.85	0.89	0.92	0.94	0.96	0.95	0.94	0.92	0.86		
Portfolio formation months are $t-48$ to $t-2$												
a	-0.73	-0.32	-0.09	-0.08	-0.05	-0.00	0.07	0.10	0.15	0.37		
b	1.16	1.12	1.06	1.05	1.02	1.01	1.00	0.99	1.04	1.11		
s	1.59	0.87	0.64	0.52	0.48	0.42	0.41	0.40	0.42	0.49		
h	0.90	0.60	0.44	0.44	0.36	0.31	0.18	0.11	-0.05	-0.26		
$t(a)$	-2.91	-2.79	-0.96	-0.99	-0.67	-0.01	1.08	1.46	2.09	3.60	2.02	0.031
$t(b)$	18.61	39.22	46.55	53.19	57.82	63.78	64.72	58.62	57.02	43.37		
$t(s)$	17.91	21.36	19.68	18.61	19.17	18.51	18.52	16.61	16.22	13.40		
$t(h)$	8.91	12.94	11.93	13.78	12.61	11.87	7.34	4.19	-1.55	-6.35		
R^2	0.73	0.88	0.91	0.92	0.93	0.94	0.95	0.93	0.94	0.90		
Portfolio formation months are $t-60$ to $t-13$												
a	-0.18	-0.16	-0.13	-0.07	0.00	0.02	0.06	0.10	-0.07	-0.12		
b	1.13	1.09	1.07	1.04	0.99	1.00	1.00	1.01	1.06	1.15		
s	1.50	0.83	0.67	0.59	0.47	0.38	0.35	0.40	0.45	0.50		
h	0.87	0.54	0.50	0.42	0.34	0.29	0.23	0.13	-0.00	-0.26		
$t(a)$	-0.80	-1.64	-1.69	-0.99	0.02	0.40	0.96	1.43	-0.92	-1.36	1.29	0.235
$t(b)$	20.24	44.40	55.03	61.09	63.79	65.68	62.58	58.26	60.49	53.04		
$t(s)$	18.77	23.63	24.09	24.06	21.21	17.44	15.43	16.18	18.06	16.33		
$t(h)$	9.59	13.67	15.94	15.31	13.46	11.82	8.98	4.46	-0.14	-7.50		
R^2	0.75	0.91	0.93	0.94	0.94	0.94	0.94	0.93	0.94	0.93		

Why momentum breaks it

Losers load *high* on s and h — like small, distressed value stocks — so the model predicts *high* returns. But momentum losers earn *low* returns: the sign flips.

The verdict

$t-12:t-2$: $GRS = 4.45$, $p = 0.000$ — *rejected*. Longer horizons recover: $t-48:t-2$ ($p = 0.031$) and $t-60:t-13$ ($p = 0.235$) are largely captured.

Interpretation: multifactor minimum variance

The three factors aren't sacred — any three spanning portfolios do the same job.

What “it prices returns” means

The factors are (close to) **multifactor minimum-variance** (MMV): minimum variance for a given expected return *and* given factor loadings.

Just as in mean–variance math any two frontier portfolios span the frontier, *any three* MMV portfolios generate every other.

So S, B, H, L need not be “true” factors — given how well SMB and HML price average returns, they're close enough.

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ICAPM: factor 1 is the market; the other two proxy for shifting investment opportunities.

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Why SMB and HML specifically?

They're the *least correlated* construction, which makes the loadings easy to interpret — not because they're uniquely “the” factors.

FF show you need at least three (Table VIII), then that any three work (Table IX).

Any three spanning portfolios do the job

The factors are interchangeable combinations of one another — the algebra behind “any three.”

						R^2	$s(e)$	
S	=	0.28 (1.99)	+1.17 M (36.95)	+ e		0.79	2.68	
L	=	-0.10 (-1.15)	+1.20 M (62.84)	+ e		0.92	1.62	
H	=	0.46 (4.08)	+0.99 M (38.73)	+ e		0.80	2.16	
SMB	=	0.19 (1.32)	+0.21 M (6.54)	+ e		0.10	2.74	
HML	=	0.56 (4.42)	-0.21 M (-7.53)	+ e		0.13	2.41	
S	=	0.00 (0.17)	-0.83 M (-29.12)	+1.00 L (46.81)	+0.81 H (50.12)	+ e	0.99	0.65
L	=	-0.03 (-0.90)	+0.86 M (51.83)	+0.86 S (46.81)	-0.67 H (-29.30)	+ e	0.99	0.60
H	=	0.06 (1.36)	+0.98 M (31.38)	+1.09 S (50.12)	-1.05 L (-29.30)	+ e	0.98	0.75
M	=	0.00 (0.08)	-0.85 S (-29.12)	+1.03 L (51.83)	+0.75 H (31.38)	+ e	0.98	0.66

FF (1996): factor spanning regressions.

What it shows

Top rows: S , L , H on the market alone leave large residual variation ($R^2 = 0.79\text{--}0.92$).

Bottom rows: once the others are included, each factor is spanned almost perfectly ($R^2 = 0.98\text{--}0.99$).

Takeaway

The three-factor space is what matters, not the particular SMB/HML construction.

The scorecard: any three spanning portfolios, same errors

Whichever three spanning portfolios you pick, the average pricing errors are the same.

Dependent Ports.	Explanatory Ports.		GRS	p(GRS)	Ave a	Ave a ²	Ave R ²
25 Size-BE/ME	M		2.76	0.000	0.286	0.1140	0.77
25 Size-BE/ME	M	SMB HML	1.97	0.004	0.093	0.0164	0.93
25 Size-BE/ME	M	S H	2.06	0.002	0.097	0.0170	0.93
25 Size-BE/ME	M	S L	2.16	0.001	0.102	0.0183	0.92
25 Size-BE/ME	M	L H	1.87	0.008	0.094	0.0159	0.92
25 Size-BE/ME	S	L H	2.06	0.002	0.094	0.0162	0.92
E/P	M		2.85	0.002	0.260	0.1059	0.83
E/P	M	SMB HML	0.84	0.592	0.051	0.0039	0.93
E/P	M	S H	0.95	0.488	0.059	0.0051	0.94
E/P	M	S L	1.02	0.427	0.064	0.0057	0.94
E/P	M	L H	0.86	0.575	0.052	0.0041	0.93
E/P	S	L H	0.86	0.571	0.051	0.0040	0.93
Sales Rank	M		2.51	0.006	0.256	0.0821	0.82
Sales Rank	M	SMB HML	0.87	0.563	0.053	0.0058	0.93
Sales Rank	M	S H	1.01	0.437	0.055	0.0068	0.94
Sales Rank	M	S L	0.96	0.474	0.052	0.0059	0.94
Sales Rank	M	L H	0.92	0.514	0.052	0.0057	0.93
Sales Rank	S	L H	0.93	0.509	0.052	0.0057	0.93
C/P & Sales Rank	M		2.93	0.002	0.268	0.1007	0.80
C/P & Sales Rank	M	SMB HML	1.04	0.405	0.062	0.0068	0.93
C/P & Sales Rank	M	S H	1.13	0.338	0.067	0.0068	0.93
C/P & Sales Rank	M	S L	1.14	0.333	0.063	0.0064	0.93
C/P & Sales Rank	M	L H	1.03	0.416	0.061	0.0064	0.92
C/P & Sales Rank	S	L H	1.05	0.396	0.061	0.0065	0.93
60-13	M		2.51	0.006	0.268	0.0899	0.80
60-13	M	SMB HML	1.29	0.235	0.092	0.0114	0.92
60-13	M	S H	1.38	0.186	0.094	0.0112	0.92
60-13	M	S L	1.19	0.299	0.077	0.0074	0.92
60-13	M	L H	1.29	0.234	0.089	0.0102	0.91
60-13	S	L H	1.30	0.230	0.090	0.0107	0.91
12-2	M		5.13	0.000	0.337	0.1647	0.79
12-2	M	SMB HML	4.46	0.000	0.331	0.2097	0.90
12-2	M	S H	4.45	0.000	0.322	0.2027	0.90
12-2	M	S L	4.58	0.000	0.329	0.2040	0.90
12-2	M	L H	4.51	0.000	0.326	0.2047	0.90
12-2	S	L H	4.46	0.000	0.328	0.2069	0.90

The MMV signature

For each test set, the five three-factor specifications (M+SMB+HML, M+S+H, M+S+L, M+L+H, S+L+H) give nearly identical average |a| (circled) — any three spanning portfolios price the cross-section equally.

The exception

Momentum (12-2) is rejected under every specification ($p = 0.000$). No three-factor span captures it.

Is it actually risk? The missing piece

The model works. Whether that makes HML a risk factor is a separate, harder question.

Points in favor

HML captures sizeable common covariation in returns.

There are common factors in *fundamentals* too — earnings and sales, not just prices.

It absorbs a host of anomalies, including the finer LSV double-sorts.

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The unfinished business

To call HML *risk*, you must still:

identify the **state variable** behind it (FF nominate *distress* — possibly tied to investors' human capital), and

explain why it carries a *special premium* — not just covary with the economy.

A better market proxy, or a non-hedging factor, could explain it away.

FF push back on the behavioral reading

If HML were pure mispricing, two things should look different than they do.

Argument 1: the Sharpe ratio

The Sharpe ratio on HML is *not* especially high, and HML stays volatile. *H* has underperformed *L* about as often as small has underperformed big, and as often as the market has underperformed the riskless rate. That looks like a risk factor, not a money pump.

Argument 2: the timing

The distress premium persists for *at least 5 years*. But the mean reversion in earnings growth that LSV lean on happens much *sooner*. A premium that outlasts the supposed expectation error is hard to square with the overreaction story.

And a caveat on their own side

FF concede distress is *not* a macro variable — so its correlation with GNP or the market is expected to be low. Convenient for the data, awkward for the “risk” label.

Could the anomalies just be data mining?

FF check the usual ways a result can be spurious — and largely clear them.

Survivor bias?

No — the factors are *value-weighted* and built only from NYSE stocks, which limits the small-distressed-survivor problem.

Data snooping?

Probably not — Davis (1994) finds the effect pre-1962, and there is a body of *international* evidence. Out-of-sample tests, not the same sample re-mined.

The cleanest defense

One model captures essentially *all* the anomalies (save momentum). If there were many independent flukes, no single three-factor model could line them up. The unification is itself evidence against pure data mining.

Still on the table

A *poor market proxy* could be doing some of the work — the true market portfolio is unobservable (Roll's critique never fully goes away).

The puzzle FF leave open: momentum

Not data mining, not captured, and a problem for both camps.

Why it won't go away

Not data mining — confirmed out of sample (Jegadeesh–Titman 2000; Rouwenhorst 1999).

Not captured by the three-factor model.

A challenge to *behavioral* finance too (Barberis–Shleifer–Vishny 1999; Daniel–Hirshleifer–Subrahmanyam 1998; Hong–Stein 1999 all try).

Could it simply be *another factor*?

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Where this leaves the three-factor model

One anomaly, unexplained, sitting next to a model that explains everything else. Either the model is missing a factor — or “explaining anomalies” is easier than it looks.

**Daniel & Titman
(1997):
characteristics or
covariances?**

A more fundamental question

Is the value premium paid for covariances (how you co-move) or for characteristics (what you are)?

What DT ask

Are there *pervasive factors* tied to size and BE/ME?

Are there *risk premia* on the loadings?

Their answer to both: **no** — returns track the *characteristic*, not the covariance.

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Whose theory is on trial

Covariances priced by a factor is exactly the *risk-based* (FF) prediction. The *behavioral* (LSV) story says nothing about covariances vs. characteristics — it only claims cheap stocks are mispriced. So DT's test is really a test of the **risk-based** explanation, not the behavioral one.

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The key claim

High-BE/ME stocks *do* co-move — but because they share *characteristics* (industry, region), not because they load on a distress factor. They were already correlated *before* becoming distressed: the comovement predates the supposed risk.

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A timely backdrop

And the premium is far from constant — *growth beat value over 1990–2000*. A risk premium *can* stay negative for a decade, which is exactly why settling the covariance-vs-characteristic question matters.

Three return-generating hypotheses

DT lay out the rivals precisely, so the data can choose between them.

Risk model 1 — distress factor, constant premia

$$\mathbb{E}[r_{i,t}] = r_{f,t} + \sum_k \beta_{i,k} \lambda_k + \theta_{i,t-1} \lambda_D$$

BE/ME proxies for the distress loading $\theta_{i,t-1}$; a real distress factor carries its own premium λ_D .

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Non-risk model 3 — characteristics

$$\mathbb{E}[r_{i,t}] = a + b_1 \theta_{i,t-1}$$

Returns depend on the *characteristic* θ directly, with *no* link to the covariance structure. The radical option: it implies an *asymptotic arbitrage* — identical risk, different characteristics, different returns.

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The strategy

Test 1 kills model 1. Test 2 separates model 2 from model 3.

Test 1: does risk change when distress does?

If BE/ME marks a distress factor, a firm's covariation should shift as it becomes distressed.

Pre- vs. post-formation standard deviations (Table II)

Compare a portfolio's return SD *before* and *after* it becomes high- or low-BE/ME.

Result: not much difference.

If distress were a switched-on risk exposure, variance should jump. It doesn't.

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A fair worry

Does BE/ME actually *change* enough across formation to give this test power? DT have to defend that the sort really does move firms in and out of distress.

Test 2: hold the characteristic, vary the loading

The clean experiment — change factor exposure while keeping the characteristic fixed.

The double sort

First sort stocks on *characteristics* (size and BE/ME).

Within each characteristic group, sort again on **HML loading**.

Use *pre-formation* loadings (estimated $t-42$ to $t-7$) and portfolios to tame the errors-in-variables problem.

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Model 2 vs. model 3

Model 2 (covariances price): within a characteristic group, higher HML loading $\Rightarrow$ higher return.

Model 3 (characteristics price): return is flat across loadings once the characteristic is fixed.

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Sample

All NYSE, AMEX, and NASDAQ firms, 1973–1993.
Pre-formation loadings successfully rank post-formation loadings — so the sort is real, not noise.

Covariances don't capture expected returns

Across HML-loading quintiles, holding the characteristic fixed, average returns barely move.

Char Port		Factor Loading Portfolio				
BM	SZ	1	2	3	4	5
Small growth → 1	1	0.202	0.833	0.902	0.731	0.504
1	2	0.711	0.607	0.776	0.872	0.710
? → 1	3	0.148	0.287	0.396	0.400	0.830 ✓
2	1	1.036	0.964	1.014	1.162	0.862
2	2	0.847	0.957	0.997	0.873	0.724
2	3	0.645	0.497	0.615	0.572	0.718
3	1	1.211	1.112	1.174	1.265	0.994
3	2	1.122	1.166	1.168	1.080	0.955
Large value → 3	3	0.736	0.933	0.571	0.843	0.961
Average		0.740	0.817	0.846	0.866	0.806

DT (1997), Table III — annotations are Toby's.

Read the bottom (Average) row

Average across loading quintiles: 0.740, 0.817, 0.846, 0.866, 0.806 — *flat*. With the characteristic (BM, SZ) fixed, a higher distress *loading* earns no extra return. *Characteristics win*.

Does this make value irrational?

Not so fast — characteristics beating measured covariances isn't the same as proving mispricing.

Why the result is fragile

The characteristic could *proxy* for some true, unknown factor that the loading on *measured* HML captures poorly.

Betas are estimated *with error*, which blurs the beta–return relation; characteristics are measured *without* error, so they may win mechanically.

If loadings are *time-varying*, do past loadings even predict future ones?

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The underlined row: $BM = 1$, $SZ = 3$ — *large growth*

β_{HML} quintile	1	2	3	4	5
Return (%/mo)	0.148	0.287	0.396	0.400	<u>0.830</u>

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Large, liquid **growth** stocks are exactly where betas are measured *most precisely* — and there the return rises **monotonically** with the HML loading, a spread of 8.2% a year. Where the covariance is measured well, it *does* price returns.

The replies: not so fast, Daniel and Titman

Two influential rebuttals targeted the test — and Daniel and Titman answered back.

Davis, Fama & French (2000)

Extend the window to **1927–1997**. The DT (1997) result *disappears* over the longer sample — the characteristic/covariance gap looks like a feature of the short 1973–1993 period, not a law.

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Sorting *first* on characteristics and *then* on loadings is a *dependent* sort that biases the test toward *failing to reject* the characteristic model. The fix: use **independent** sorts.

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Where that leaves the test

Clever enough to force a sharp question — and contested enough that the methodology itself became the next battleground.

Why this fight matters beyond academia

If Daniel and Titman are right, a lot of standard practice is built on sand.

If characteristics price returns. . .

Performance evaluation: benchmarking on factor *loadings* mismeasures skill — you should benchmark on characteristics.

Cost of capital: a firm's discount rate depends on what it *is*, not just its betas.

Event studies: expected-return models built on loadings give the wrong counterfactual.

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The open question

Loadings and characteristics are tightly entangled in the data, and they only come apart in carefully constructed — and contestable — sorts.

Two decades on, the profession still hasn't fully settled which one you're actually paid for.

Where the debate stands

One robust fact, three readings of it, and one anomaly that troubles everyone.

LSV (1994)

Behavioral. Value is a mistake — extrapolation and overreaction. Survives because arbitrage is slow.

FF (1996)

Rational. Value loads on a distress factor (HML). The premium is a risk price — but the state variable is still unnamed.

DT (1997)

Characteristics. Neither factor is needed — returns track what stocks *are*. Contested by DFF (2000) and Berk (2001).

The thread to pull next

Value sometimes underperforms growth by a lot, and for a long time. How does *that* fit the risk-based vs. behavioral explanation? Next we take a deep dive into what the value premium has actually done from the 1990s to today — and try to understand it better, including how you would talk about it to investors.

Same trade, two stories: how value is sold

The academic debate has a commercial twin — and the two camps built rival asset managers.

Fama & French → Dimensional (DFA)

The *risk-based* pitch. Value is **compensation for risk** — a premium you earn for bearing distress. The message to investors: stay disciplined and diversified, hold the factor through the lean years, and you are *paid* to bear what others won't.

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Same stocks, opposite explanations

The two firms run value strategies that are roughly **0.98 correlated** — they buy nearly the same stocks. The disagreement isn't *what* to hold; it's *why* it works. The story you tell investors shapes whether they stay invested when value lags.

Value, 1990–today: the great drawdown and its recovery

A parting fact to chew on

Since 1990, *growth* has beaten value. Does that support FF (a risk premium that, like any, can go quiet for a decade) — or LSV (the mistake got arbitrated once it was published) — or something else entirely?

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Four readings of a factor that stops working

- **It was never there.** Data mining — the in-sample result was luck.
- **It was there and got arbitrated.** Publication taught everyone the trade; capital competed the premium away.
- **It is there, and this is what risk feels like.** A genuine premium *must* have long, painful stretches — or no one would be paid to bear it.
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How you'd tell them apart

- **Other markets, other eras.** Data mining should fail everywhere else; a real premium should survive out of sample.
- **Valuations.** Arbitrage *compresses* the spread; a re-pricing *widens* it — that is how you separate “gone” from “cheap.”
- **Size vs. the strategy's own volatility.** Is the bad run outside what a 0.4-Sharpe strategy should produce, or exactly what it should?

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The uncomfortable part

All four readings look identical while you are living through them. The rest of this deck works the value drawdown as a real case.

The longest, deepest drawdown on record

From 2017 to 2020, systematic value had its worst run in roughly a century.

What happened

Value fell hard and kept falling — the deepest and longest value drawdown in the modern record. It was not one bad year; it was a multi-year grind that bottomed in 2020.

Source: AQR (Asness et al.), *Is (Systematic) Value Investing Dead?* (2020); figures are hypothetical backtests.

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Not one measure, not one shop

Over the drawdown, every standard value measure fell by roughly half:

Fama–French HML (US): about –50%

Russell Value–Growth: about –55%

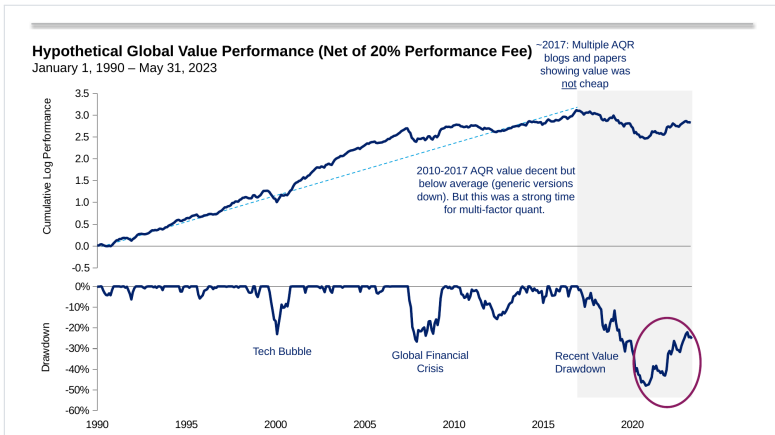
MSCI Value–Growth: about –54%

AQR's value sleeve: about –50%

Source: AQR (Asness et al.), *Is (Systematic) Value Investing Dead?* (2020); figures are hypothetical backtests.

The drawdown, in one picture

Cumulative value (top) and its drawdown (bottom) — the deepest, longest fall on record.



Hypothetical global value, 1990–2023. Source: AQR (Asness et al.); hypothetical backtest.

“Absolute return” doesn’t mean “works all the time”

A good long-run premium still has terrible stretches — and a 0.4-Sharpe strategy is expected to.

The arithmetic of patience

The US market’s own Sharpe ratio is about 0.4. At $SR = 0.4$ you underperform cash about **34% of years** and **10% of decades**. A multi-year value drawdown is statistically ordinary — not proof the premium is gone.

Value still works on average

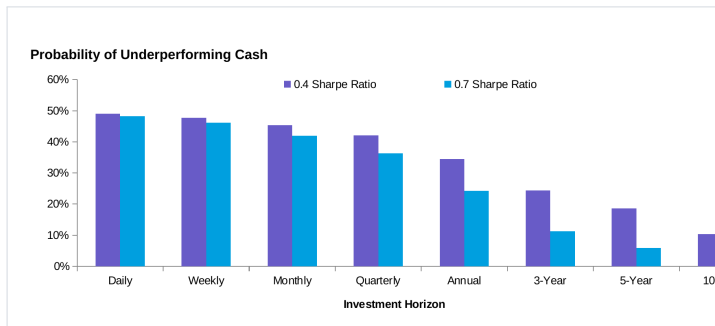
Even simple value has paid off across US, Japanese, European, and EM equities, and in bonds and currencies. One historic drawdown in one place isn’t a verdict on the premium.

The real question

Is *this* time different — did something structural finally break value, or is it just an unusually deep draw of a premium that still pays?

A 0.4-Sharpe strategy underperforms cash all the time

Even over 10 years, a 0.4-Sharpe premium underperforms cash about 10% of the time.



Probability of underperforming cash by horizon, 0.4 vs. 0.7 Sharpe. Source: AQR, Bloomberg.

What broke value? The suspects

AQR put every popular “value is dead” explanation on trial — because each one is testable.

Structural stories

Cheap stocks *deserve* to be cheap (worse fundamentals, quality)

Low interest rates and QE

Mega-caps / FANGs / a “winner-take-all” economy

Intangible assets mismeasure book value

Market-structure stories

Share buybacks distort the metrics

Crowding — too many quants arbitrated it away

We’re simply *measuring* value wrong

Usually asserted, rarely tested

Each of these is an empirical claim. So test them — one at a time.

It wasn't the usual suspects

One by one, the structural stories fail to explain the drawdown.

Ruled out

Not FANGs: drop the largest 10%, the tech/media/telecom industries, or the most-expensive decile — the spread barely moves.

Not intangibles: drop the high-intangible names — still historically wide.

Not deserved cheapness: profitability, ROA, margins, and 5-yr earnings-growth gaps between cheap and expensive were within their normal range.

Also ruled out

Not recessions or crashes: value shows little sensitivity to NBER recessions or the 10 worst market drawdowns.

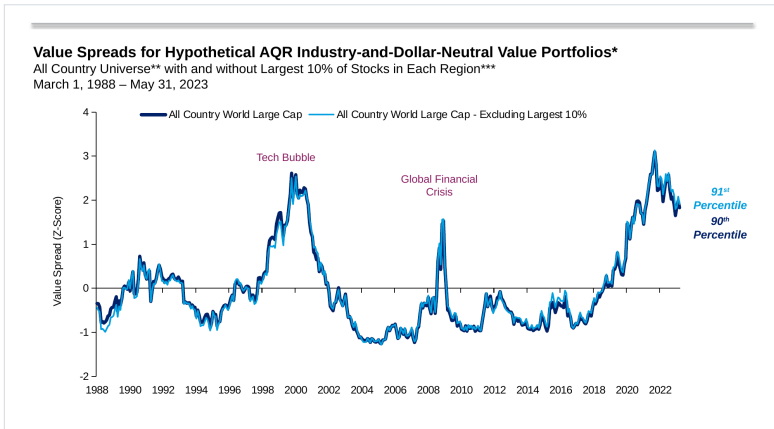
Mostly not rates: the recent value–rate correlation did rise (about 0.1 → 0.5), but rates explain only 15–30% of the drawdown and recovery (Maloney & Moskowitz 2020).

What's left

Strip the structural stories away and the drawdown is overwhelmingly a *re-pricing*.

It wasn't the mega-caps

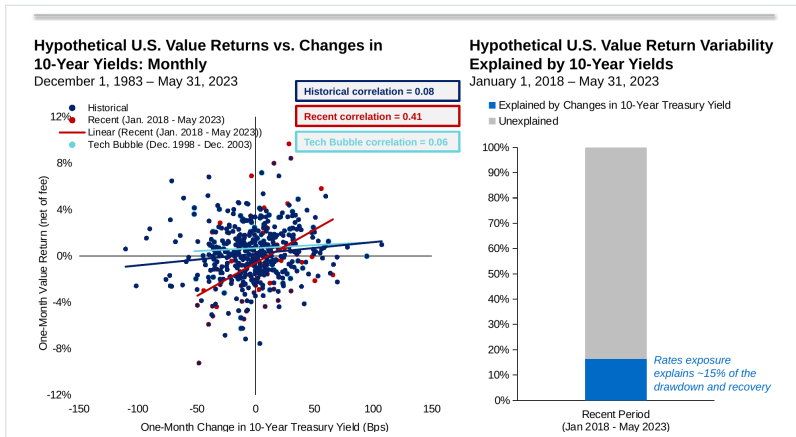
Drop the largest 10% of stocks and the spread barely moves — both near the 90th percentile.



Value spread with and without the largest 10%, 1988–2023. Source: AQR; hypothetical backtest.

And it wasn't an interest-rate bet

Correlation rose recently ($0.08 \rightarrow 0.41$), but rates explain only about 15% of the move.



Value returns vs. 10-year-yield changes, and variance explained. Source: AQR.

It was (almost) all re-pricing: the value spread

The drawdown wasn't fundamentals deteriorating — it was the price of cheapness collapsing.

What the value spread is

Sort stocks by valuation; take the ratio of the *expensive* basket's multiple to the *cheap* basket's. That ratio is the **value spread** — how much more you pay for glamour than for value.

It went to an extreme

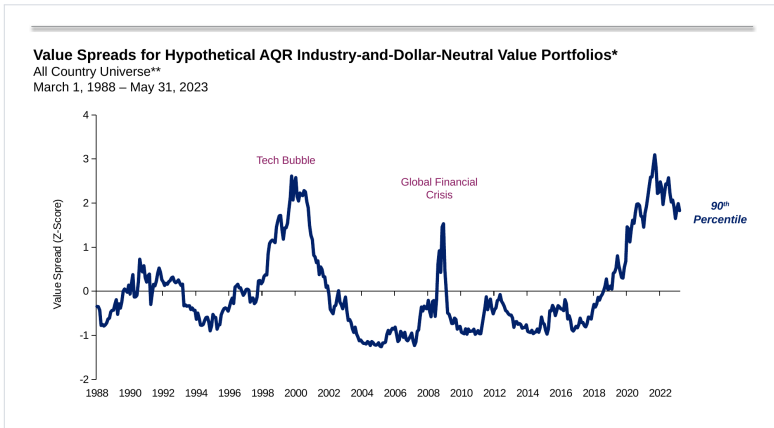
By 2020 the spread reached about the **99th percentile** of its history — *wider than the peak of the 2000 tech bubble*. Cheap had rarely, if ever, been this cheap relative to expensive.

Why the spread is the whole story

A *widening* spread *is* the drawdown — you mark value down as it gets cheaper. And a *wide* spread is the setup for the recovery: the cheaper value gets, the more it earns when the gap finally closes.

The value spread, 1988–2023

By 2020–21 the spread blew past its tech-bubble peak.



Value spread (Z-score), all-country universe. Source: AQR; hypothetical backtest.

A dislocation never seen before

At these prices, the market was betting on growth no fundamentals have ever delivered.

The implied bet

Back out the growth needed to justify the spread: at the extreme, expensive firms would have to *out-grow* cheap firms by about **80–110% over five years**.

Never realized

That gap is far larger than *any* expensive-minus-cheap growth differential ever realized in the data. Expensive firms do grow faster — but nowhere near this much.

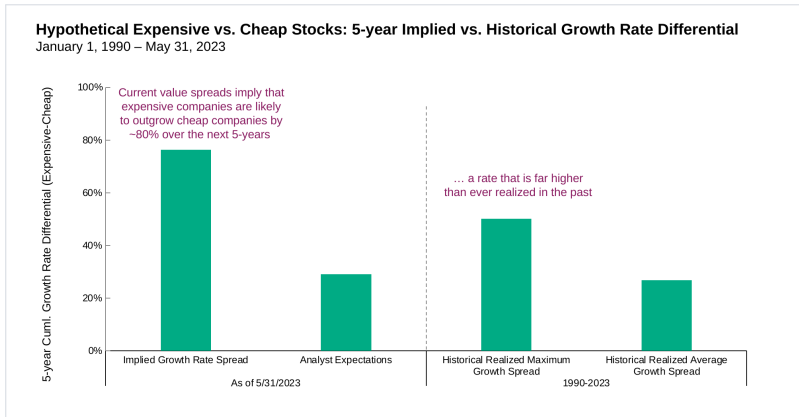
LSV (1994), live

This is the extrapolation error of Lakonishok–Shleifer–Vishny in real time: investors paying for aspirational “disruptor” growth that fundamentals never deliver.

The last time the data looked like this was **1999–2000**.

Implied growth vs. anything ever realized

Implied expensive—cheap growth far exceeds anything ever realized.



5-year cumulative expensive—cheap growth differential: implied vs. realized. Source: AQR.

The recovery

Wide spreads forecast returns — and value came back.

History rhymes

After severe dislocations, value has historically rebounded hard. Following the tech bubble (2000–2003), global value returned roughly **+75% to +108%**.

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This time

The recovery began around **November 2020** and value rallied sharply — but not in a straight line, and by 2023–24 spreads were *still* wide (80th–90th percentile). On AQR's read: “more to go.”

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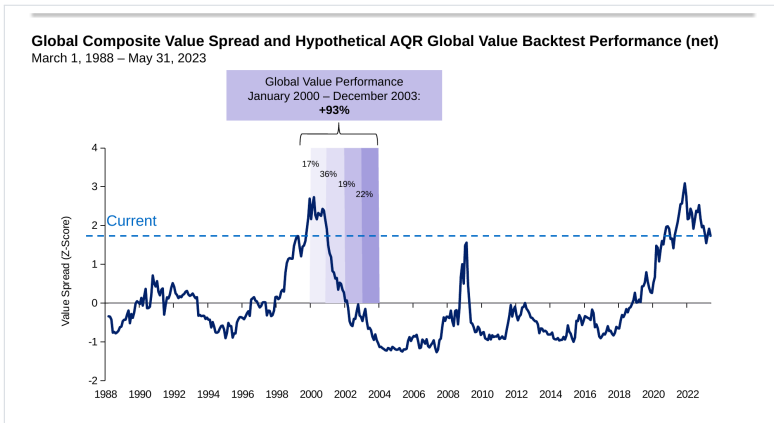
The instrument

The spread is the practical forecast: a high spread $\Rightarrow$ high expected future value returns.

It is the one variable *both* camps watch — and the closest thing to a usable signal for timing the premium's bad and good stretches.

Recoveries are large — and the spread is still wide

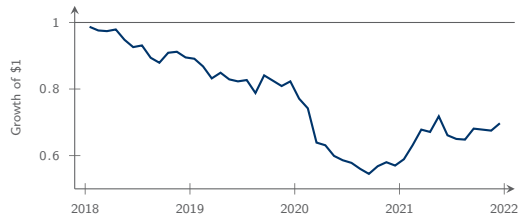
Post-tech-bubble, value returned about +93% (2000–2003); today's spread is still wide.



Value spread with the post-2000 recovery annotated; “Current” marked. Source: AQR.

The value factor since 2018: the fall, then the bounce

US HML, 2018–2021



−30.3% over the window; −44.8% peak-to-trough.

US HML, 2021–Jun 2026

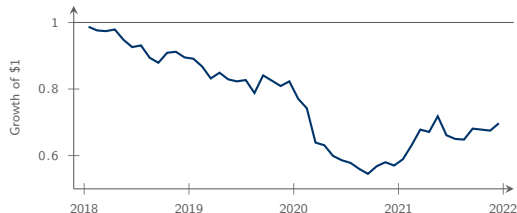


+56.1% over the window — the sharpest value rally since 2000.

Computed from Kenneth R. French's data library, July 2026 database. HML is the monthly value factor; the line is the compounded growth of \$1 in the long-short factor.

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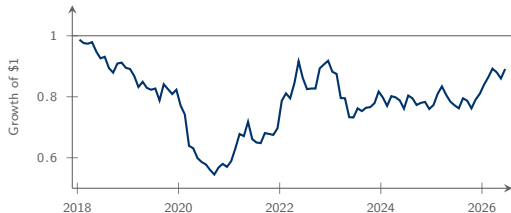
Two windows, two completely different stories

Pick your start date and value is either dead or thriving. This is precisely why a factor's track record is so easy to argue about — and why the *next* slide matters more than either of these.

Computed from Kenneth R. French's data library, July 2026 database. HML is the monthly value factor; the line is the compounded growth of \$1 in the long-short factor.

Put them together: value is *still* below its 2018 level

US HML, 2018–Jun 2026



—10.9% over the full episode. Eight and a half years, and the factor has not made it back.

US HML, 1990–Jun 2026

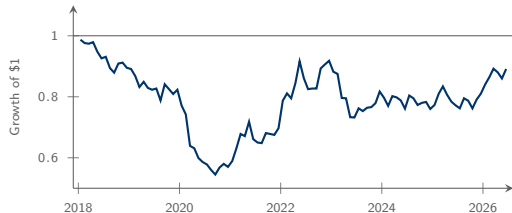


+63.4% over 36 years — about 1.4% a year, with a —57.8% drawdown along the way.

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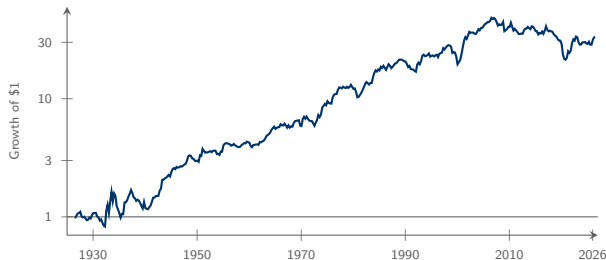
What a 36-year record looks like up close

A premium that compounds at 1.4% a year and loses more than half its value en route is not a money pump — but it is also not obviously arbitrated away. **Which reading does this picture support?**

Computed from Kenneth R. French's data library, July 2026 database. HML is the monthly value factor; the line is the compounded growth of \$1 in the long-short factor.

The full century: US HML, 1926–2026

US HML, Jul 1926–Jun 2026 (log scale)

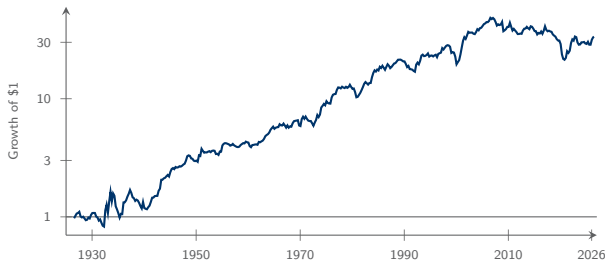


- \$1 becomes **\$33.5** — +3,249% over a century.
- But the **worst drawdown in the entire 100 years** is the recent one: **–57.8%**.
- Note the log scale: the last decade's fall is as large, in proportional terms, as anything in the record — including the Depression.

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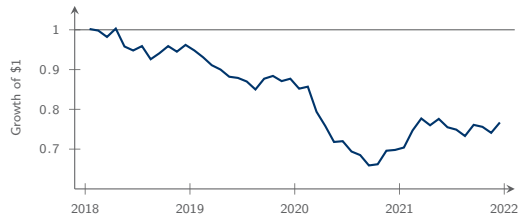
The honest summary

The premium is real over a century *and* the recent experience is genuinely unprecedented. Both things are true.

Computed from Kenneth R. French's data library, July 2026 database. HML is the monthly value factor; the line is the compounded growth of \$1 in the long-short factor.

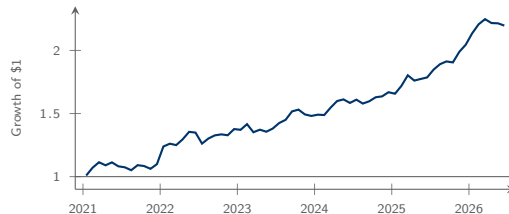
The same test abroad: developed ex-US HML

Ex-US HML, 2018–2021



−23.3%; −34.3% peak-to-trough — bad, but milder than the US.

Ex-US HML, 2021–Jun 2026

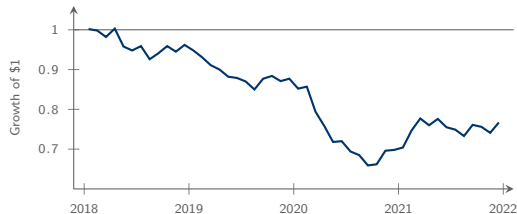


+119.7% — the factor more than doubled.

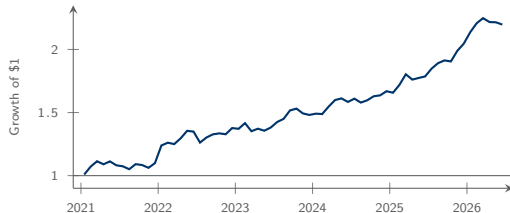
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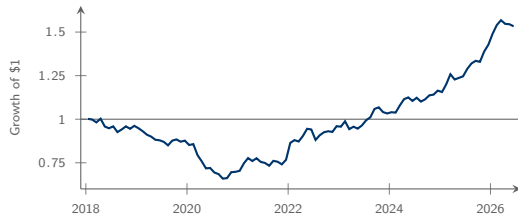
Out-of-sample, in the best sense

The drawdown showed up outside the US too, so it was not a US data quirk — and the recovery abroad has been far stronger. A factor that fails in one market and recovers in another is telling you something about *breadth*.

Computed from Kenneth R. French's data library, July 2026 database. HML is the monthly value factor; the line is the compounded growth of \$1 in the long-short factor.

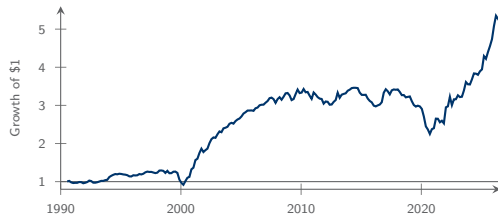
Ex-US: fully recovered, and a stronger long record

Ex-US HML, 2018–Jun 2026



+53.3% over the same episode in which US HML lost 10.9%.

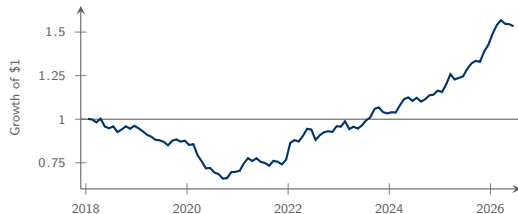
Ex-US HML, Jul 1990–Jun 2026



+423% since inception — the series starts in Jul 1990, so this *is* the full history.

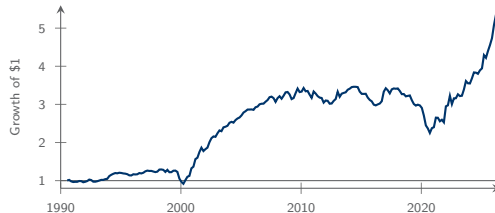
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So has value “stopped working”?

Over the identical 36 years, ex-US HML returned +423% against the US factor’s +63%. If you had judged value on the US alone you would have drawn a very different — and much gloomier — conclusion. **Ask what any single market’s track record can actually establish.**

Risk or behavioral? The episode as a live test

The same LSV-vs-FF debate — now with real money and a real –50% drawdown.

Reads as behavioral (LSV)

A re-pricing, not a fundamentals story; spreads *predict* returns; a near-replay of 1999–2000; the “mistake” visibly widened and then corrected.

Reads as risk (FF)

A genuine risk premium *can* go negative for a decade; in real time you can't tell a bad draw from a vanished premium; the spread could be a time-varying *price of risk*.

The honest answer

The drawdown's character — re-pricing, predictability, the 1999 echo — leans behavioral. But a wide spread is consistent with *either* a fattening mistake *or* a rising price of risk. The data under-determine the *why*.

How you'd talk to investors

The hard part isn't believing value works — it's holding it through a -50% drawdown.

What the episode teaches

A great long-run premium will have stretches that feel like failure.

The valuation *spread*, not recent returns, is the forward-looking signal.

The premium is earned by *staying invested* through the pain.

Two stories, one trade

DFA (FF) sells value as *risk you're paid to bear*; LSV sells it as *others' mistake you exploit*. The same ≈ 0.98 -correlated portfolio — but the story decides whether a client holds on at the bottom.

The takeaway

Whether value is a risk premium or a mistake, the investor's job is the same: size it so you can *survive* the drawdown, and lean in when the spread is wide.